



Morphological Image Processing

Part 1: Binary image

Digital Image Processing



Rafael C. Gonzalez
Richard E. Woods

Text book

- Title: Digital Image Processing
- Authors:
 - Rafael C. Gonzalez, University of Tennessee,
 - Richard E. Woods, Interapptics.
- Publisher: New York, NY : Pearson, [2018]

Objectives



Understand basic concepts of mathematical morphology, and how to apply them to digital image processing.



Be familiar with the tools used for binary image morphology, including erosion, dilation, opening, closing, and how to combine them to generate more complex tools.



Be able to develop algorithms based on binary image morphology for performing tasks such as morphological smoothing, edge detection, extracting connected components, and skeletonizing.



Be familiar with how binary image morphology can be extended to grayscale images.



Be able to develop algorithms for grayscale image processing for tasks such as textural segmentation, granulometry, computing gray-scale image gradients, and others.

Morphology

Morphology

The branch of biology that deals with the form and structure of organisms without consideration of function

Mathematical Morphology

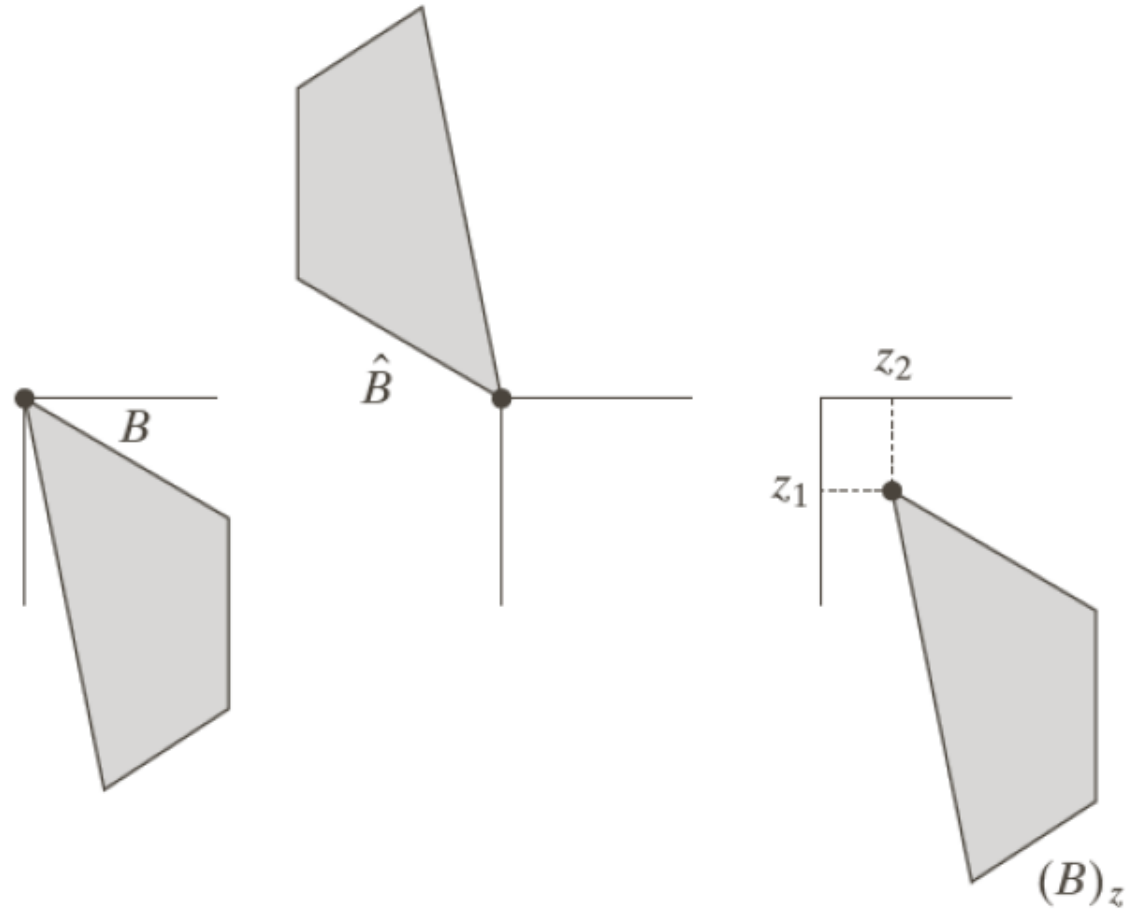
Mathematical tool for processing shapes in image, including boundaries, skeletons, convex hulls, etc.

Use of set theoretical approach

Set Theory: Definitions and Notations

- SET (Ω)
 - A collection of objects (elements)
- membership ($\in$)
 - If ω is an element (member) of a set Ω , we write $\omega \in \Omega$
- Subset ($\subset$)
 - Let A, B are two sets. If for every $a \in A$, we also have $a \in B$, then the set A is a *subset* of B , that is, $A \subset B$
 - If $A \subset B$ and $B \subset A$, then $A = B$.
- Empty set ($\emptyset$)
- Complement set
 - If $A \subset \Omega$, then its complement set $A^c = \{\omega \mid \omega \in \Omega, \text{ and } \omega \notin A\}$
- Union ($\cup$)
 - $A \cup B = \{\omega \mid \omega \in A \text{ or } \omega \in B\}$
- Intersection ($\cap$)
 - $A \cap B = \{\omega \mid \omega \in A \text{ and } \omega \in B\}$
- Set difference ($-$)
 - $B \setminus A = B \cap A^c$
 - Note that $B - A \neq A - B$
- Disjoint sets
 - A and B are disjoint (mutually exclusive) if $A \cap B = \emptyset$

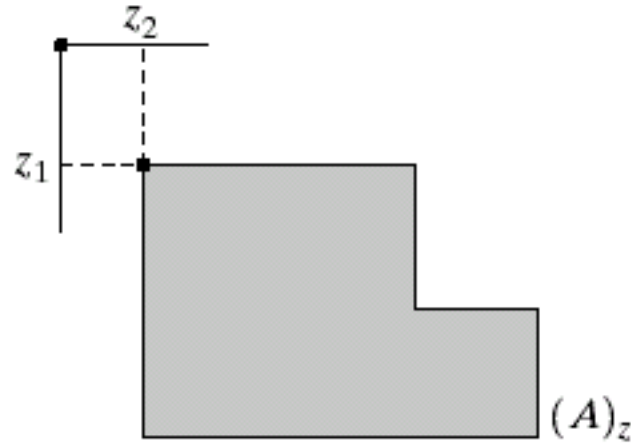
Set Relations



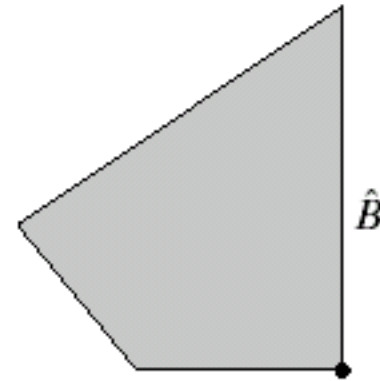
A set, its rotation, and its translation by z

Translation and Reflection

Translation of
A by z



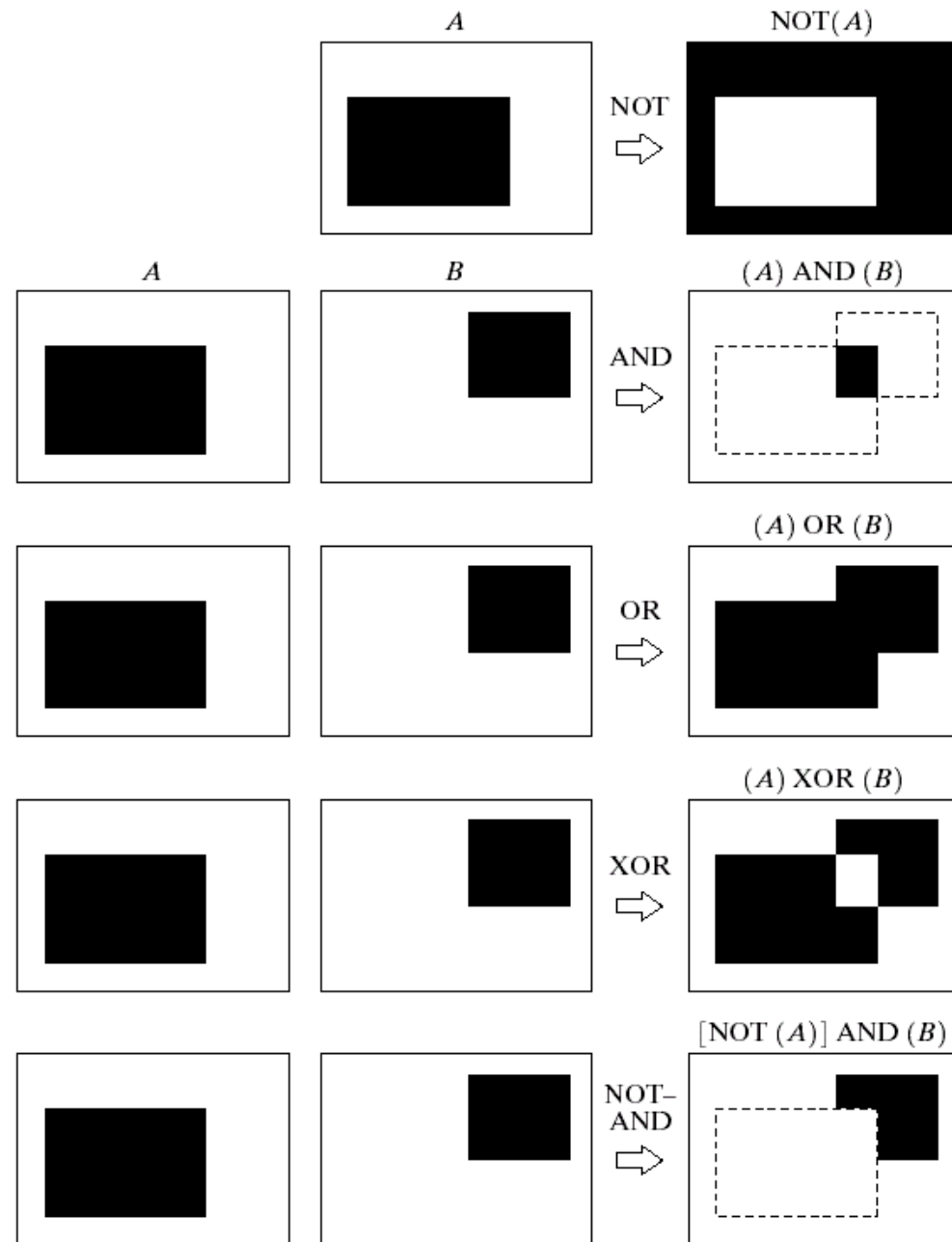
Reflection of B



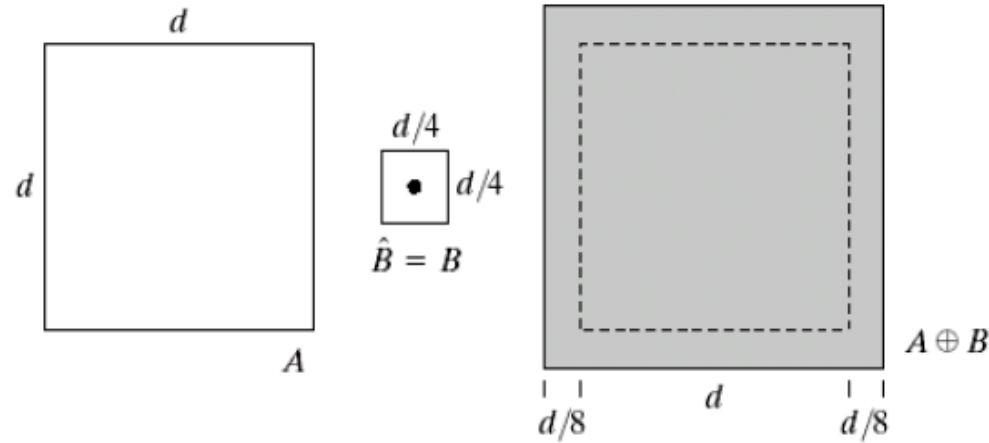
Translation $(A)_z = \{ c \mid c = a + z, \text{ for } a \in A \}$

Reflection: $\hat{B} = \{ w \mid w = -b, \text{ for } b \in B \}$

Logic Operations Between Binary Images



Dilation and Erosion

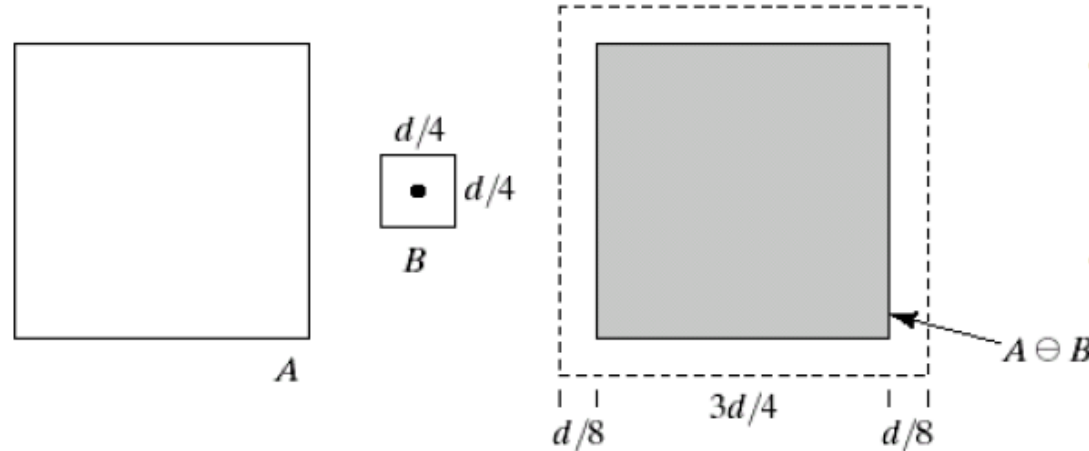


- Dilation

B : structure element

$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$$

$$= \{z \mid [(\hat{B})_z \cap A] \in A\}$$



- Erosion

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

- Relations

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

Duality Property

- Erosion and dilation are *duals* of each other with respect to set complementation and reflection
- useful when the structuring element values are symmetric with respect to its origin (as often is the case)
- we can obtain the erosion of A simply by dilating its background (i.e., dilating A^c) with the same structuring element and complementing the result.

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

$$(A \oplus B)^c = A^c \ominus \hat{B}$$

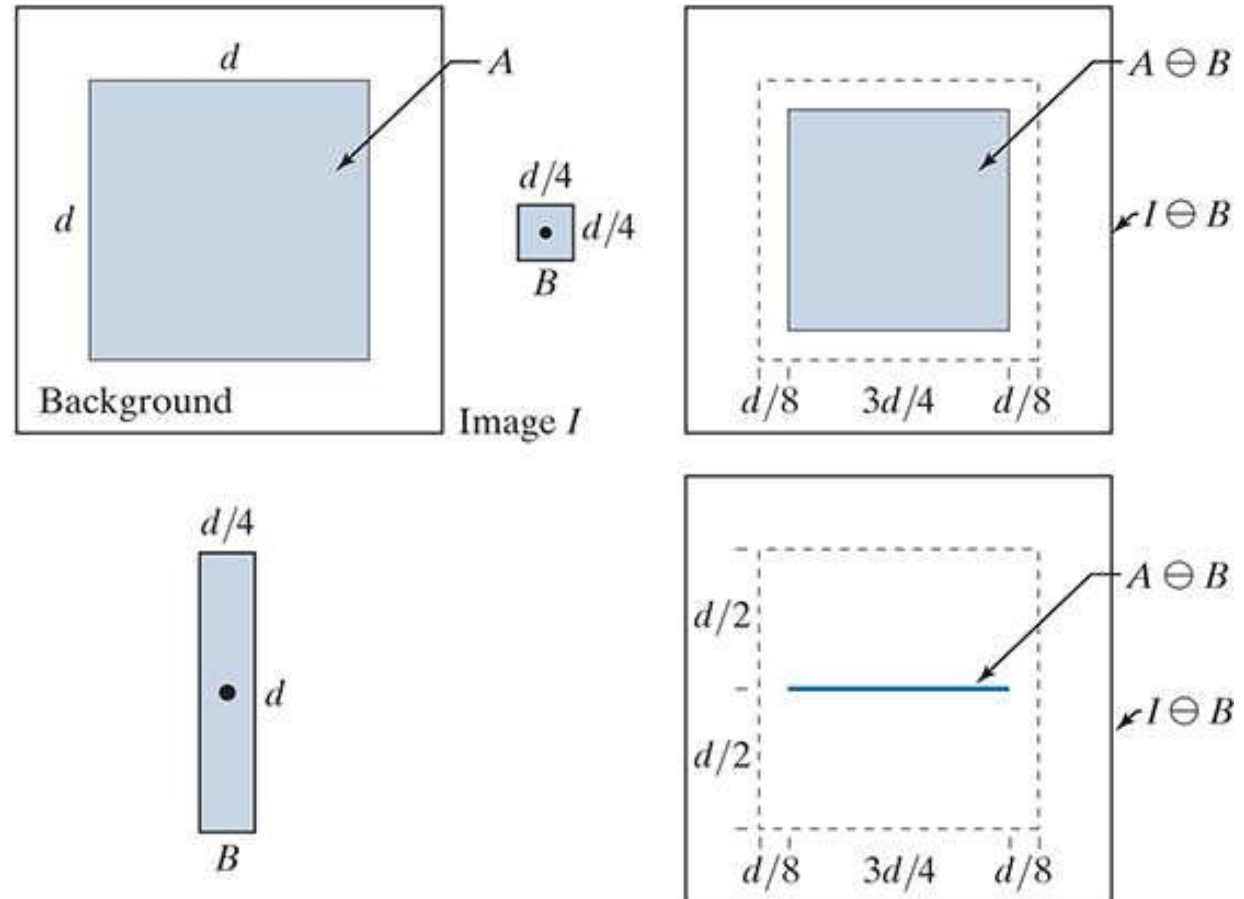
Erosion of A by B

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

a b c
d e

Figure 9.4

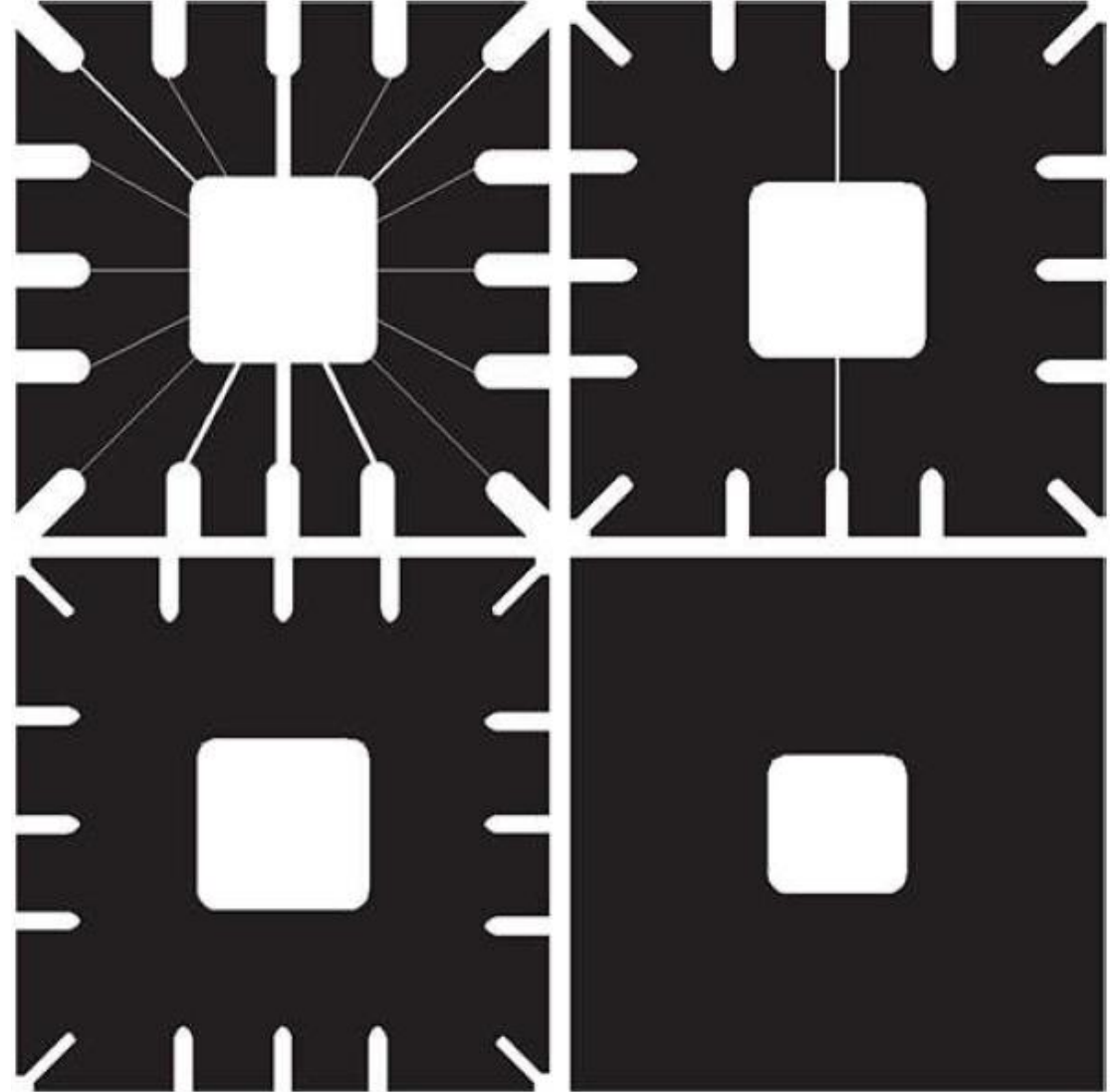
- (a) Image I , consisting of a set (object) A , and background.
- (b) Square SE, B (the dot is the origin).
- (c) Erosion of A by B (shown shaded in the resulting image).
- (d) Elongated SE.
- (e) Erosion of A by B . (The erosion is a line.) The dotted border in (c) and (e) is the boundary of A , shown for reference.



Erosion to remove image components

Figure 9.5

- (a) A 486 x 486 binary image of a wire-bond mask in which foreground pixels are shown in white.
- (b) Image eroded using square structuring elements of sizes 11 x 11
- (c) SE of size 15 x 15, and
- (d) SE of size 45 x 45 elements, respectively, all valued 1.

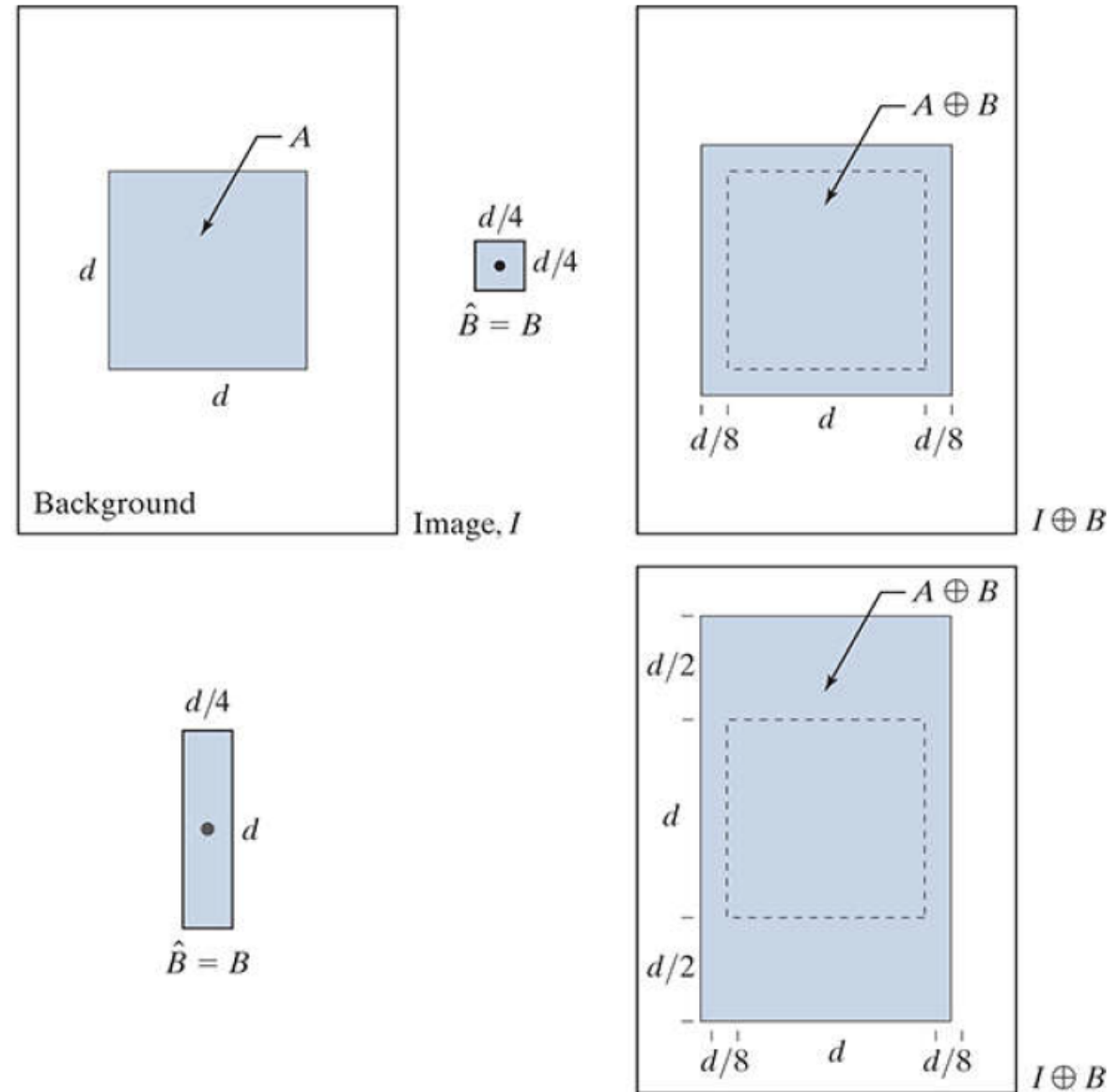


Dilation A by B

$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$$

Figure 9.6

- (a) Image I , composed of set (object) A and background.
- (b) Square SE (the dot is the origin).
- (c) Dilation of A by B (shown shaded).
- (d) Elongated SE.
- (e) Dilation of A by (d). The dotted line in (c) and (e) is the boundary of A , shown for reference.



Dilation to join broken segments

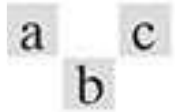
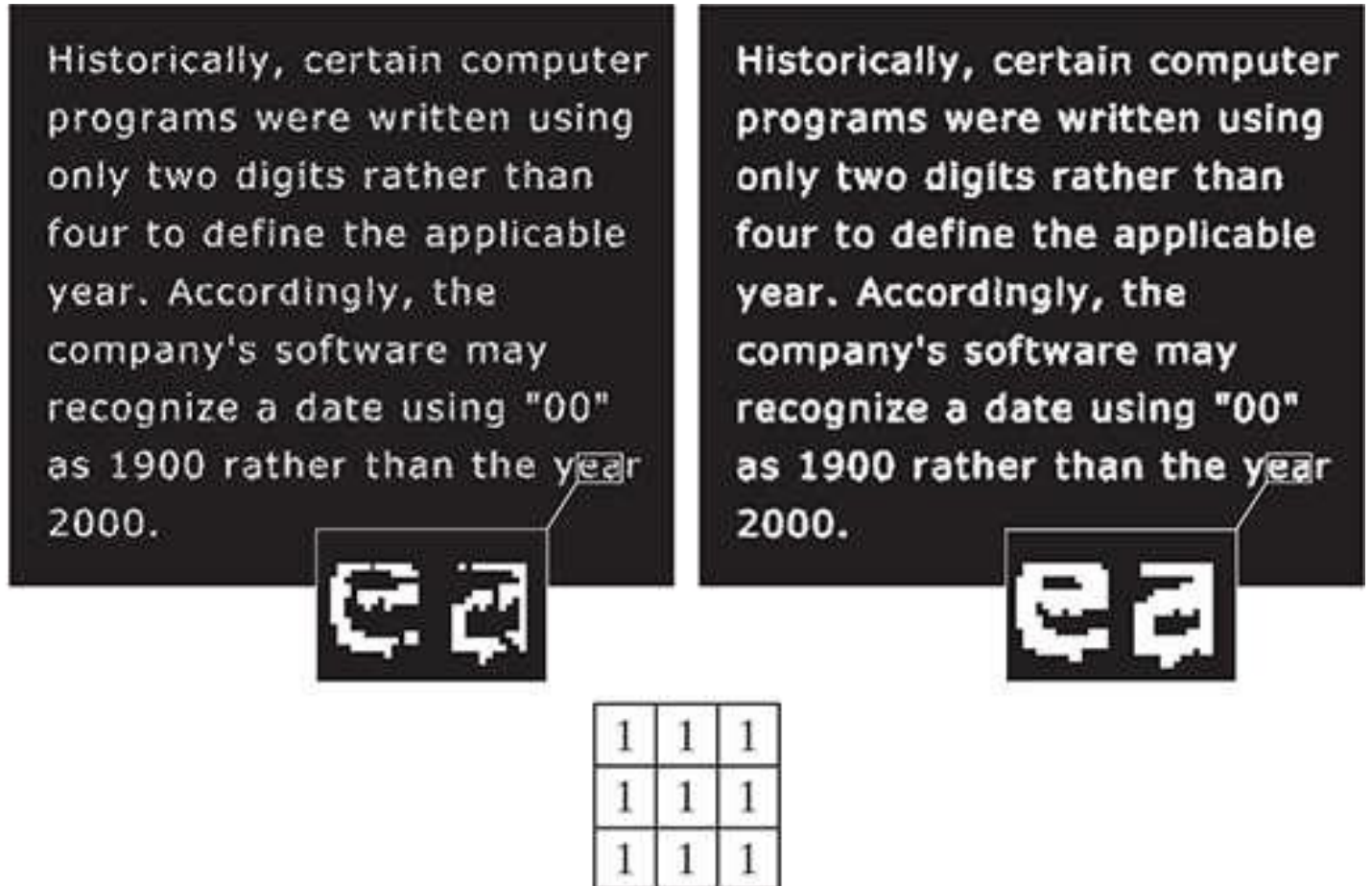


Figure 9.7

- (a) Low-resolution text showing broken characters (see magnified view).
- (b) Structuring element.
- (c) Dilation of (a) by (b). Broken segments were joined.

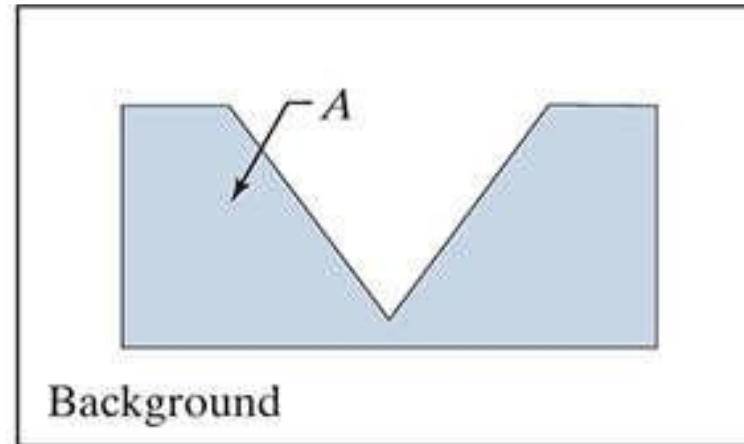


Opening

a b
c d

FIGURE 9.8

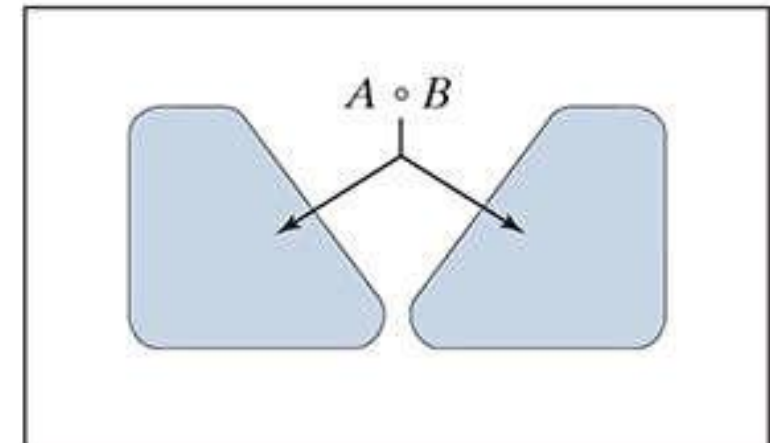
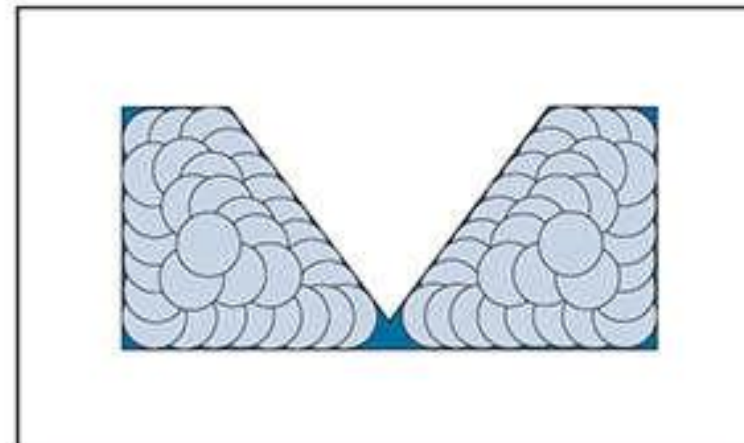
- (a) Image I , composed of set (object) A and background.
- (b) Structuring element, B .
- (c) Translations of B while being contained in A . (A is shown dark for clarity.)
- (d) Opening of A by B .



$$A \circ B = (A \ominus B) \oplus B$$



Image, I

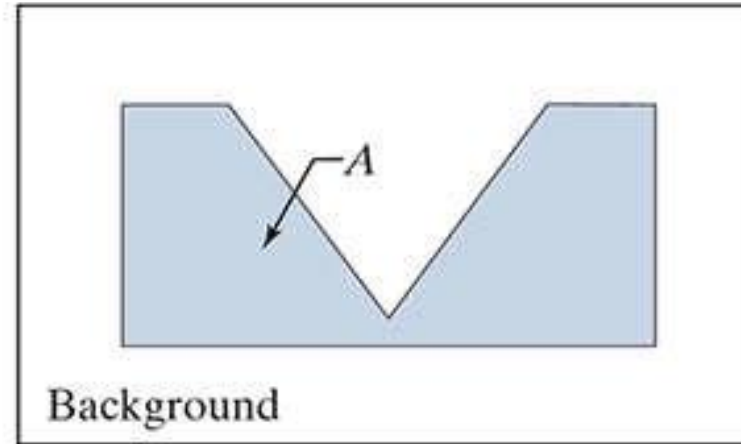


Closing

a	b
c	d

FIGURE 9.9

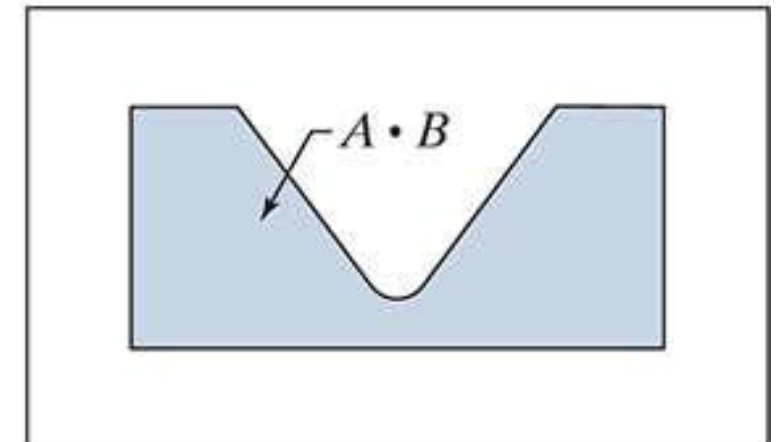
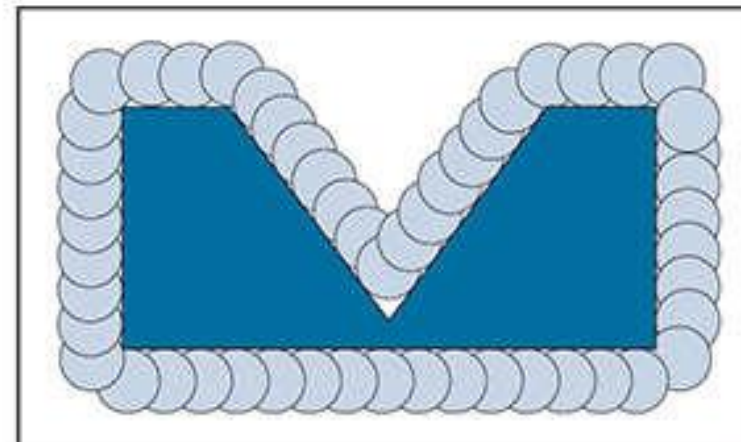
- (a) Image I , composed of set (object) A , and background.
- (b) Structuring element B .
- (c) Translations of B such that B does not overlap any part of A . (A is shown dark for clarity.)
- (d) Closing of A by B .



$$A \bullet B = (A \oplus B) \ominus B$$



Image, I



Opening and Closing

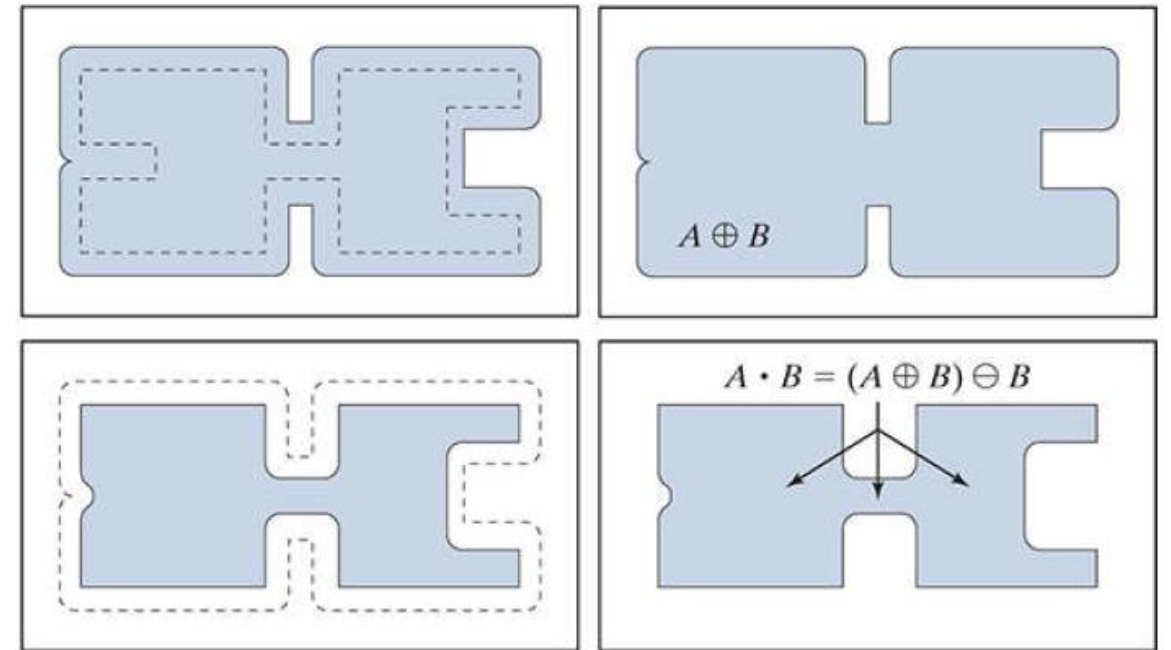
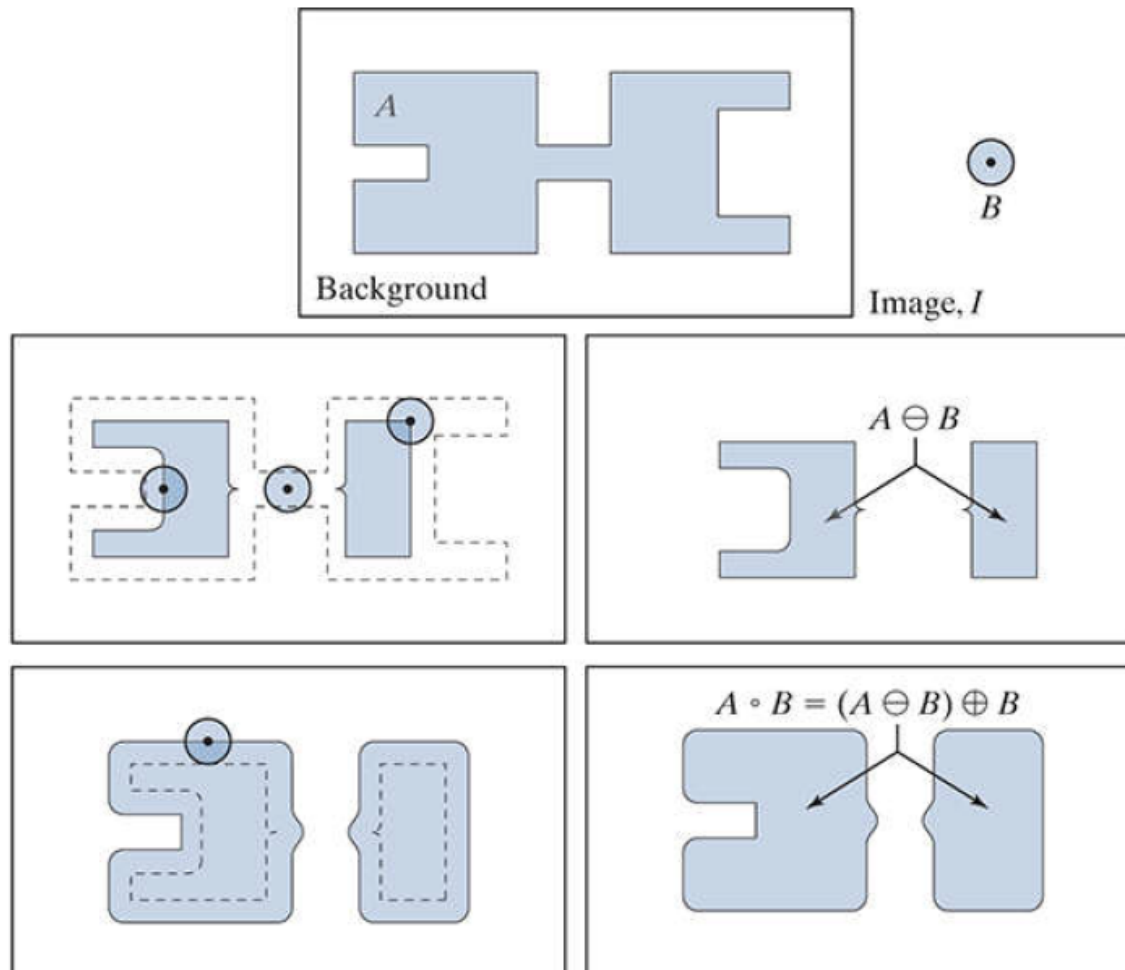


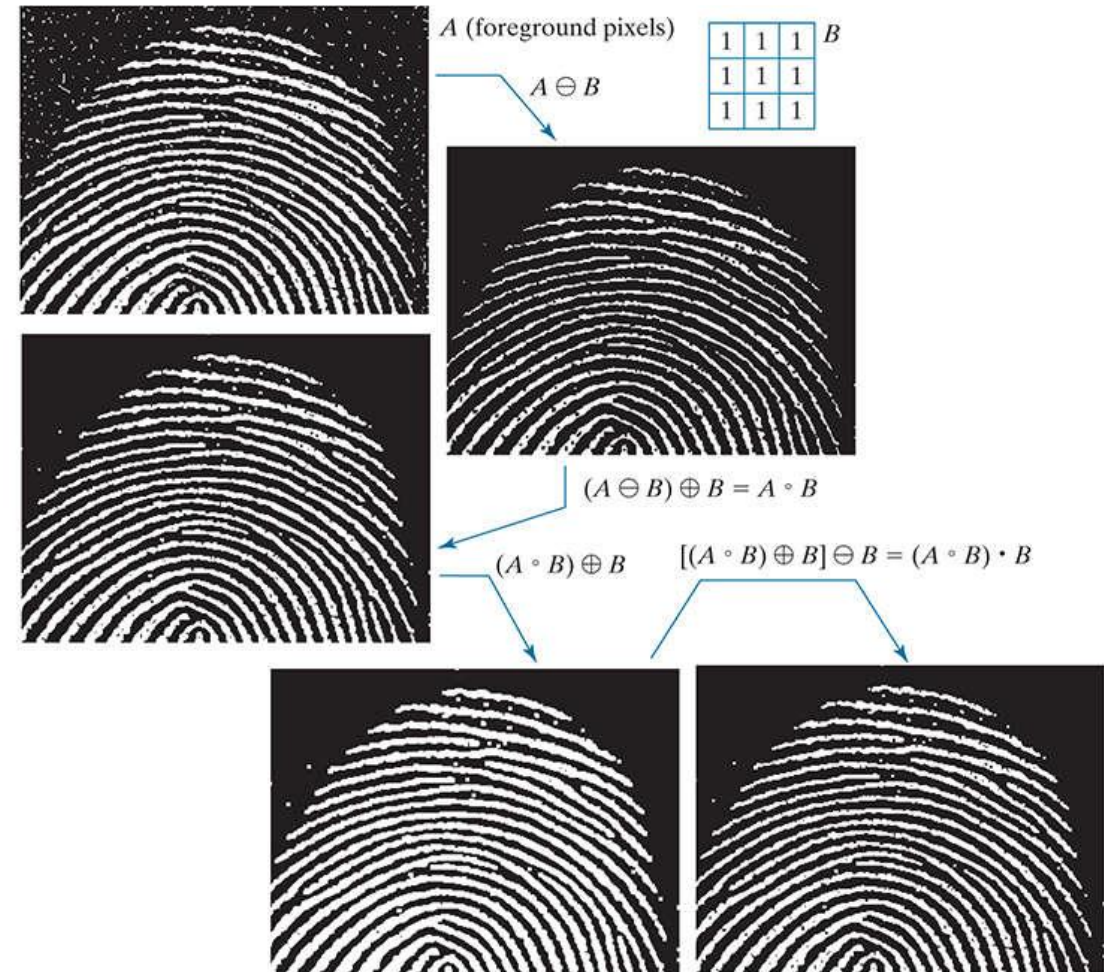
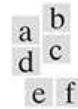
FIGURE 9.10

- (a) Image I , composed of a set (object) A and background; a solid, circular structuring element is shown also. (The dot is the origin.)
- (b) Structuring element in various positions.
- (c) - (i) The morphological operations used to obtain the opening and closing.

Opening and closing for morphological filtering

FIGURE 9.11

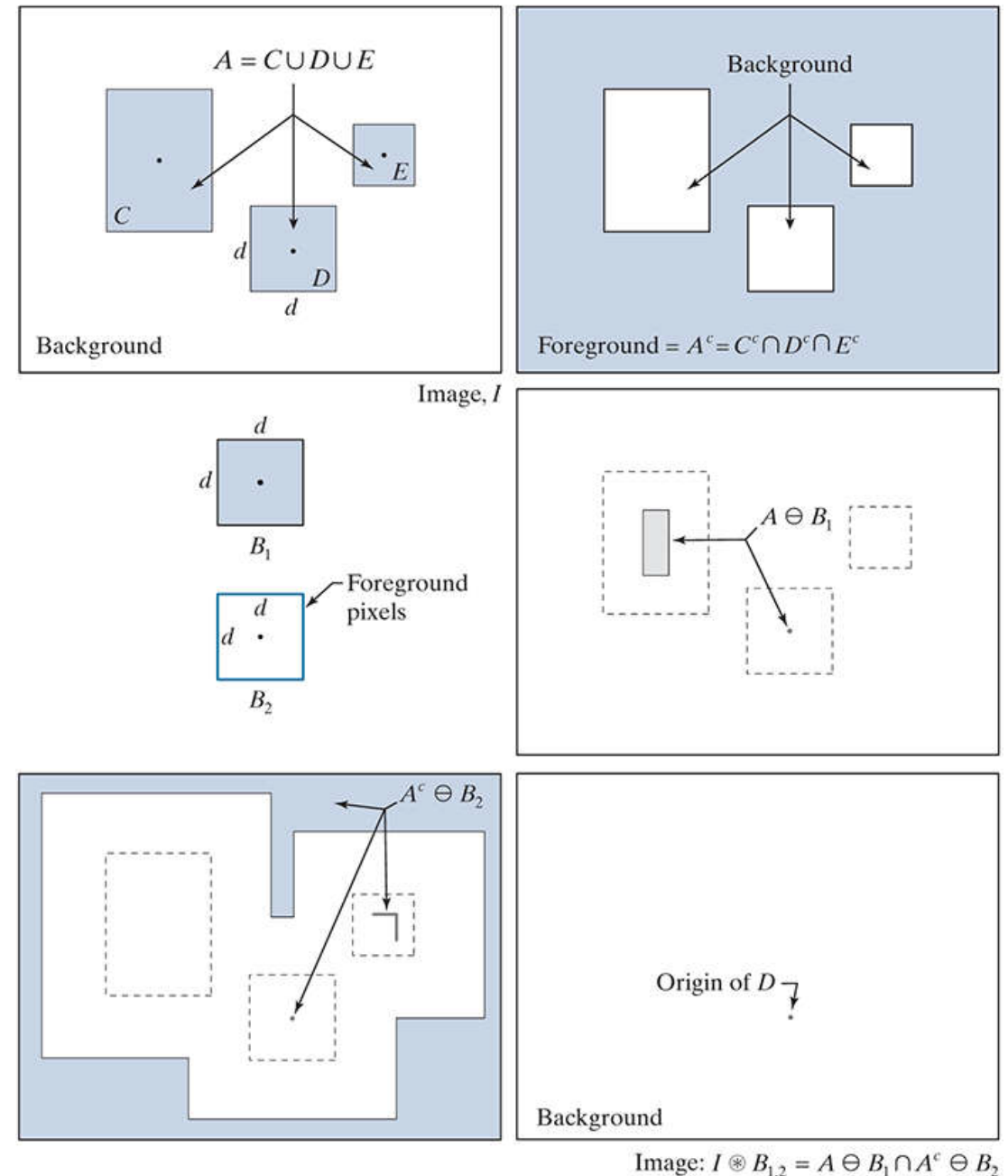
- (a) Noisy image.
- (b) Structuring element.
- (c) Eroded image.
- (d) Dilation of the erosion (opening of A).
- (e) Dilation of the opening.
- (f) Closing of the opening.



Hit-Or-Miss Transform for shape detection

FIGURE 9.12

- (a) Image consisting of a foreground (1's) equal to the union, A , of set of objects, and a background of 0's.
- (b) Image with its foreground defined as A^c
- (c) Structuring elements designed to detect object D .
- (d) Erosion of A by B_1
- (e) Erosion of A^c by B_2
- (f) Intersection of (d) and (e), showing the location of the origin of D , as desired. The dots indicate the origin of their respective components. Each dot is a single pixel.



Hit-Or-Miss Transform

- Previous example does not contain don't care entries.
- In structure element

- 1 – foreground
- 0 – background
- X – don't care

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} .* \begin{bmatrix} -1 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} : \text{match}$$

- Output is 1 if exact match of both foreground and background pixels.

- `Hitnmiss.m`
 - +1: foreground
 - -1: background
 - 0: don't care

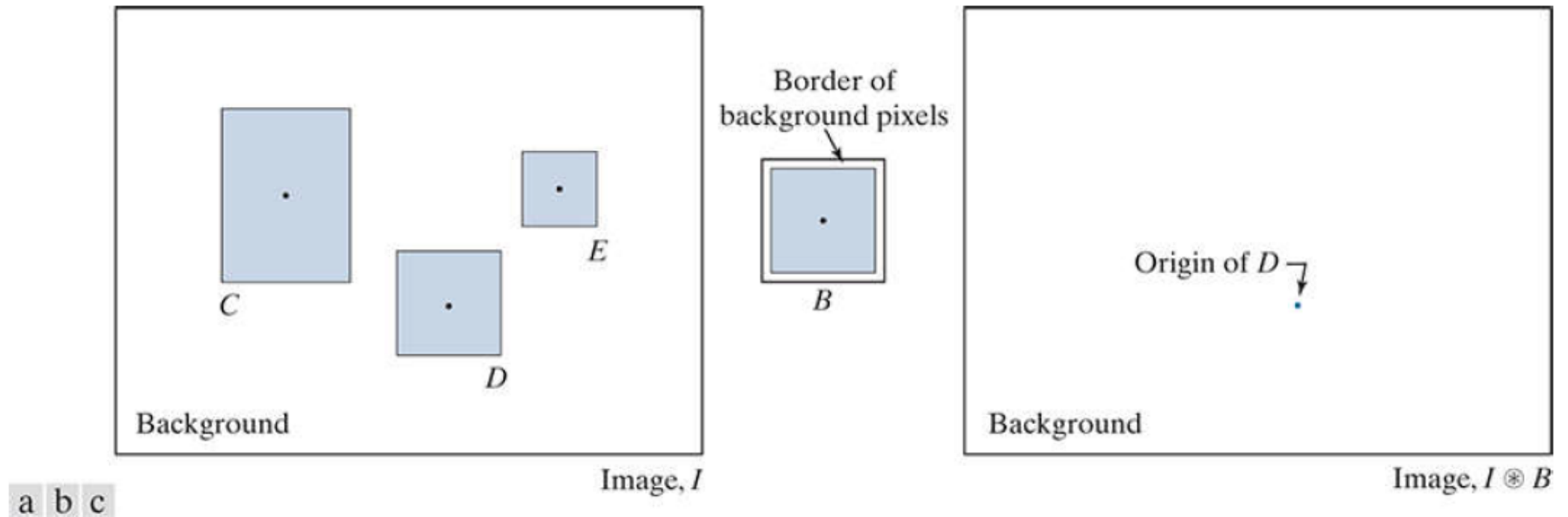
$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} .* \begin{bmatrix} -1 & -1 & -1 \\ 1 & -1 & -1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix} : \text{not match}$$

Subset image detection

Figure 9.13

Eq. 9-17 to detect subset image

$$I \circledast B = \{z \mid (B)_z \subseteq I\}$$



Detection of specific feature

FIGURE 9.14

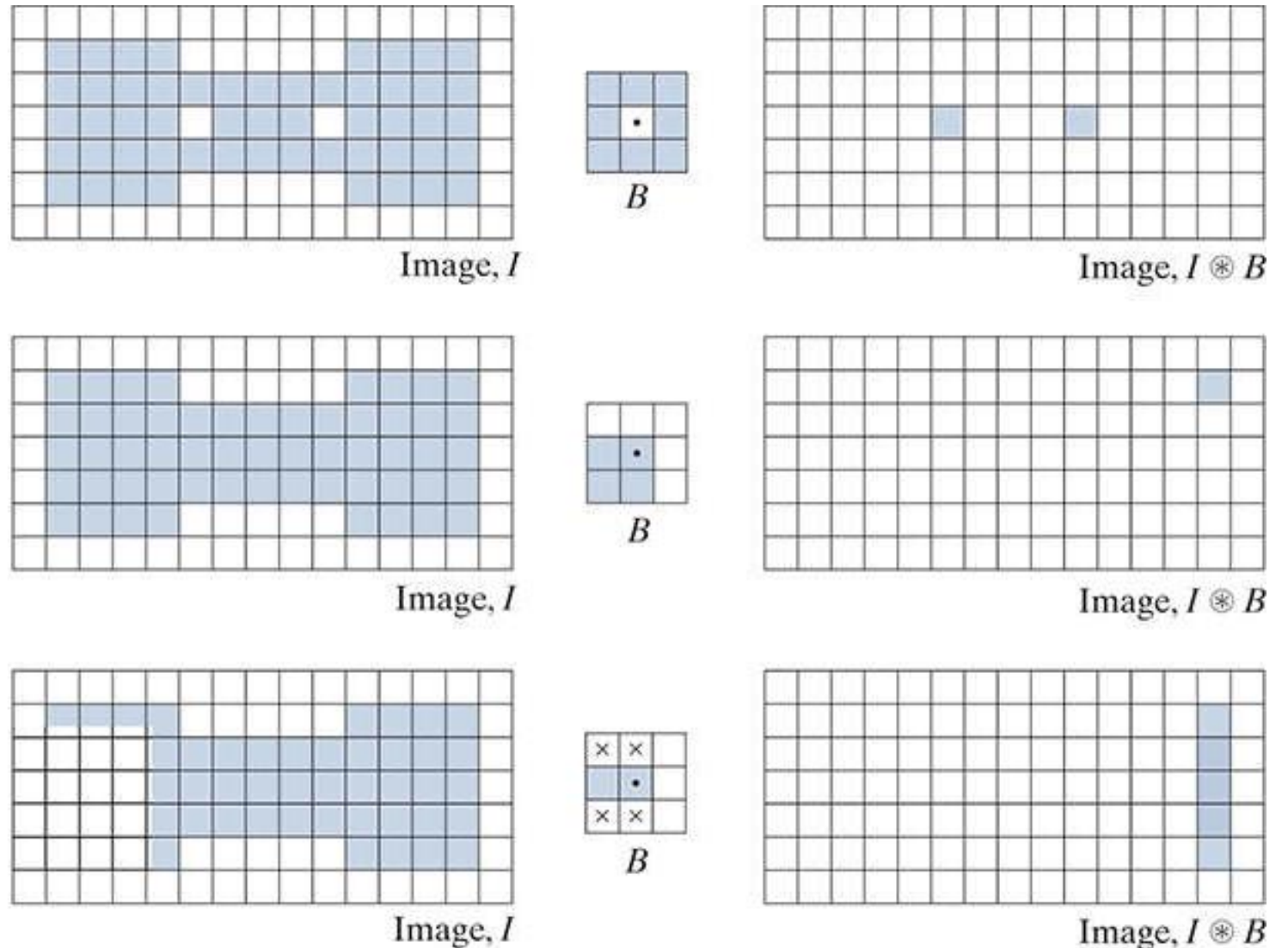
Three examples of using a single structuring element and [Eq. \(9-17\)](#) to detect specific features.

$$I \circledast B = \{z \mid (B)_z \subseteq I\}$$

First row: detection of single-pixel holes.

Second row: detection of an upper-right corner.

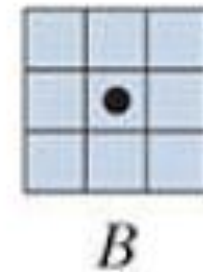
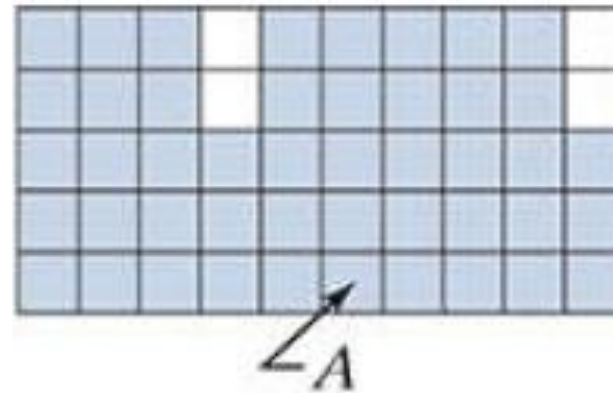
Third row: detection of multiple features.



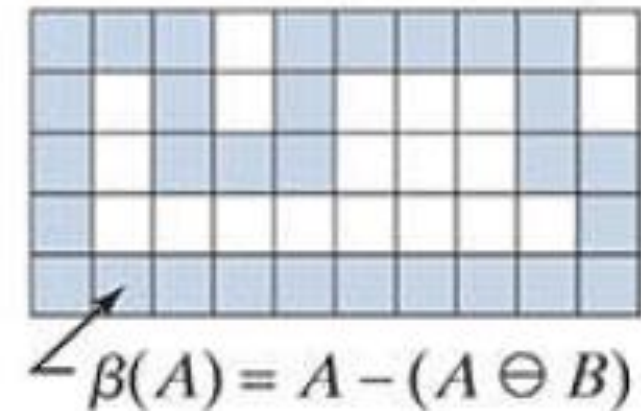
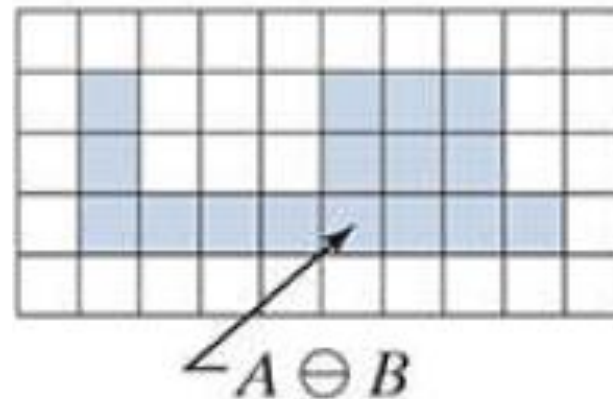
Boundary Extraction

FIGURE 9.15

- (a) Set, A , of foreground pixels.
- (b) Structuring element.
- (c) A eroded by B .
- (d) Boundary of A .



$$\beta(A) = A - (A \ominus B)$$



Boundary Extraction Example

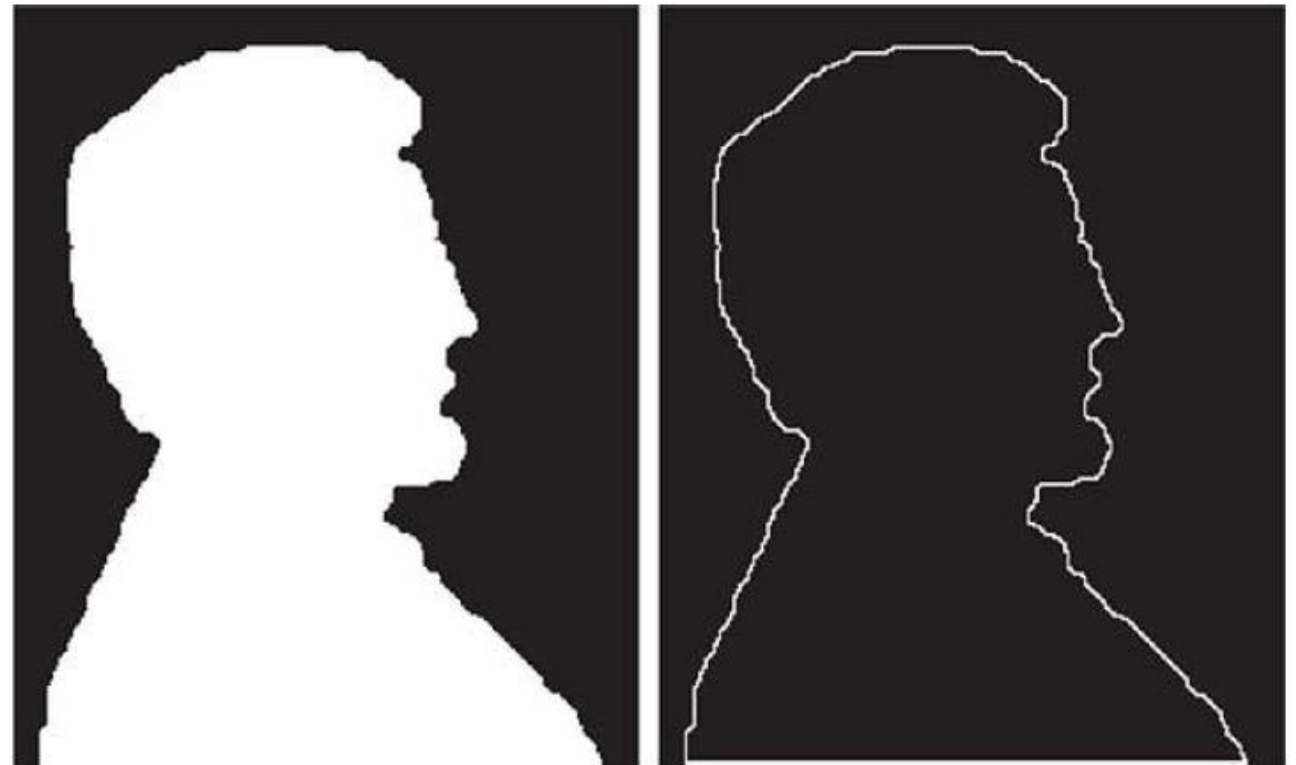
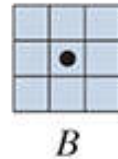
FIGURE 9.16

(a) A binary image.

(b) Result of using Eq. 9-18

$$\beta(A) = A - (A \ominus B)$$

with the structuring element B

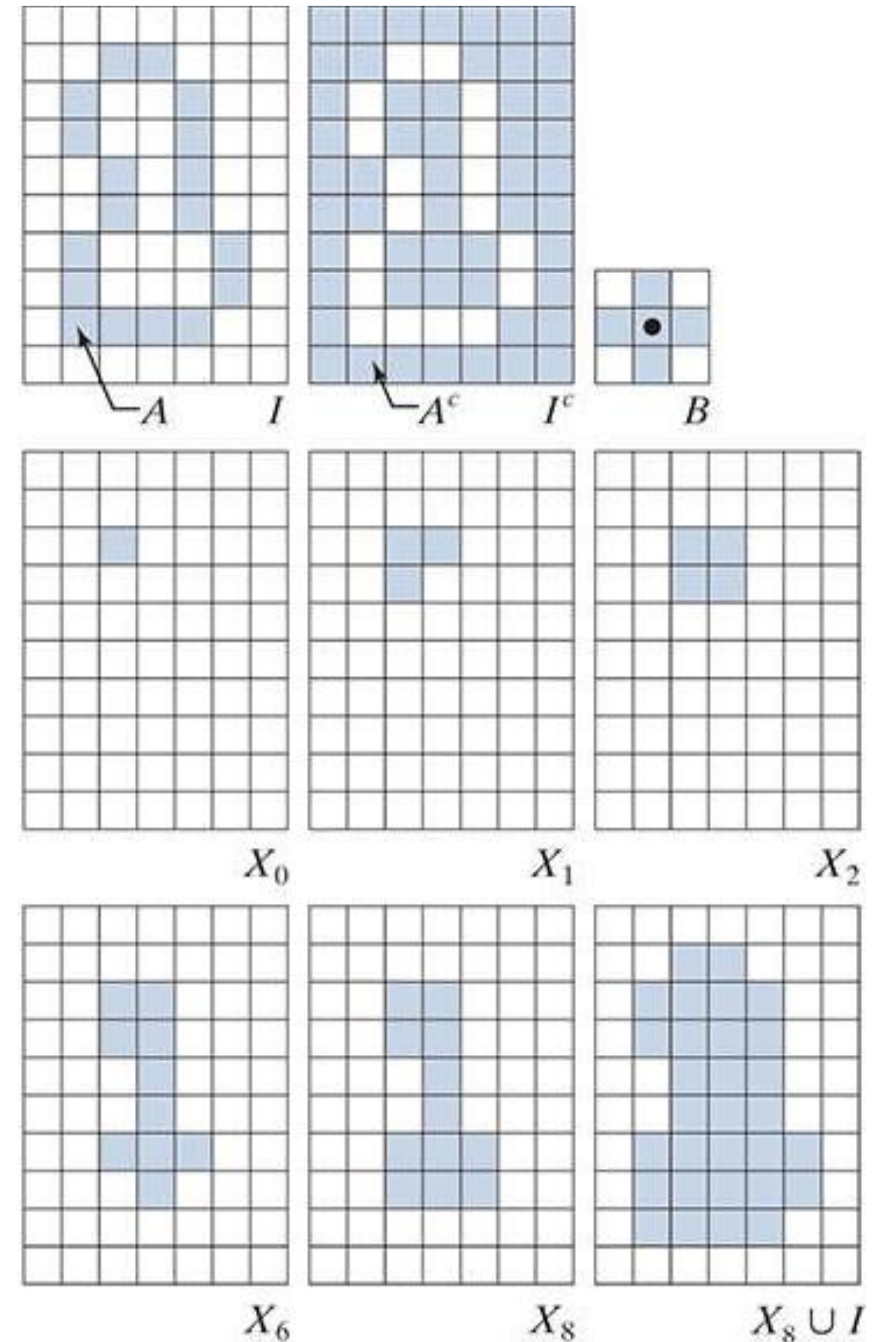


Hole Filling

Figure 9.17

- (a) Set A (shown shaded) contained in image I .
- (b) Complement of I .
- (c) Structuring element B . *Only* the foreground elements are used in computations
- (d) Initial point inside hole, set to 1.
- (e) – (h) Various steps of [Eq. \(9-19\)](#).
- (f) Final result [union of (a) and (h)].

$$X_k = (X_{k-1} \oplus B) \cap I^c \quad k = 1, 2, 3, \dots$$

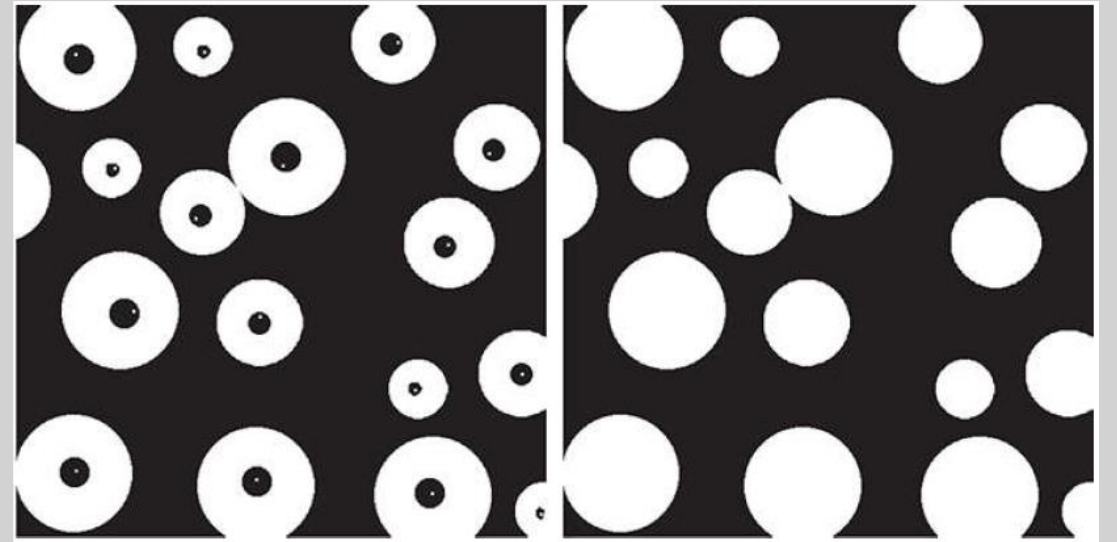


Hole Filling Example

FIGURE 9.18

(a) Binary image. The white dots inside the regions (shown enlarged for clarity) are the starting points for the hole-filling algorithm.

(b) Result of filling all holes.

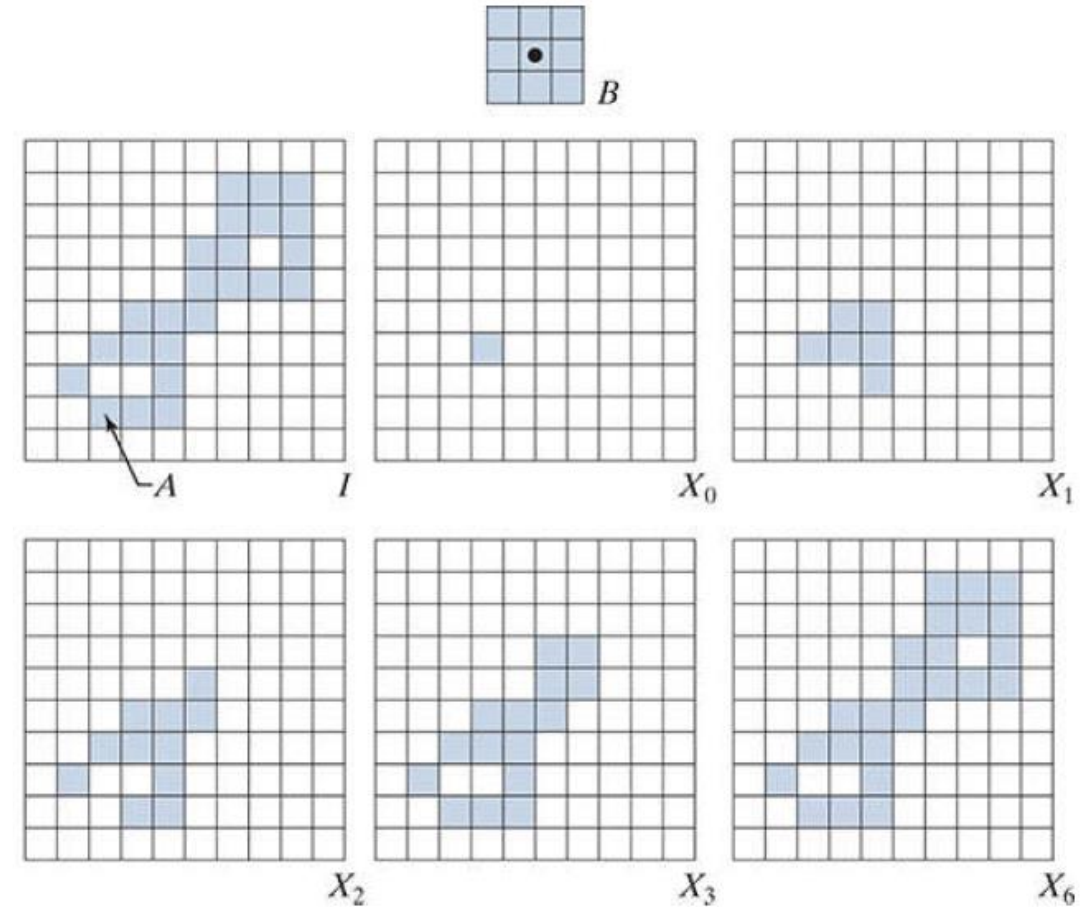


Extraction of Connected Components

FIGURE 9.19

- (a) Structuring element.
- (b) Image containing a set with one connected component.
- (c) Initial array containing a 1 in the region of the connected component.
- (d) – (g) Various steps in the iteration of **Eq. (9-20)**

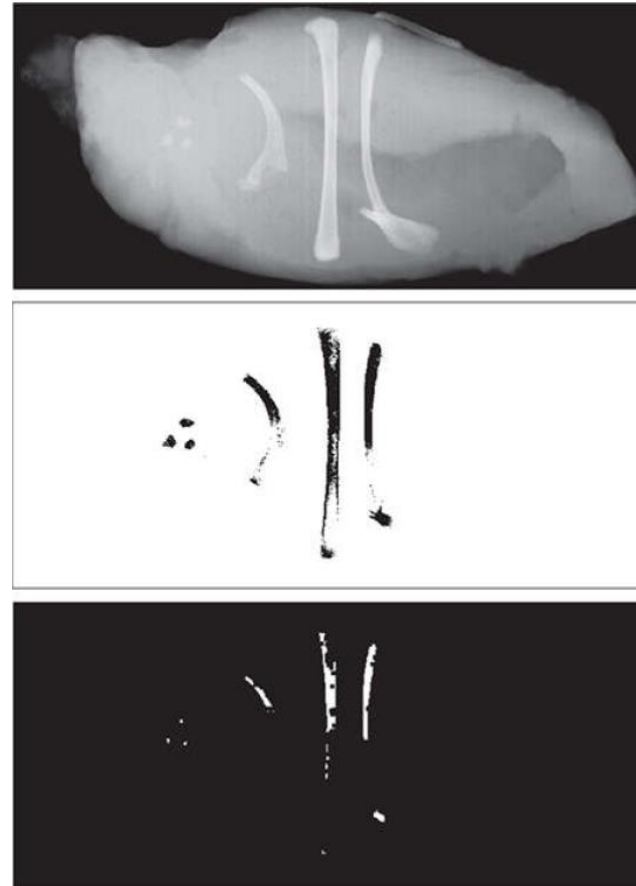
$$X_k = (X_{k-1} \oplus B) \cap I \quad k = 1, 2, 3, \dots$$



Using connected components to detect foreign objects in packaged food.

FIGURE 9.20

- (a) X-ray image of a chicken filet with bone fragments.
- (b) Thresholded image (shown as the negative for clarity).
- (c) Image eroded with a SE of 1's.
- (d) Number of pixels in the connected components of (c).



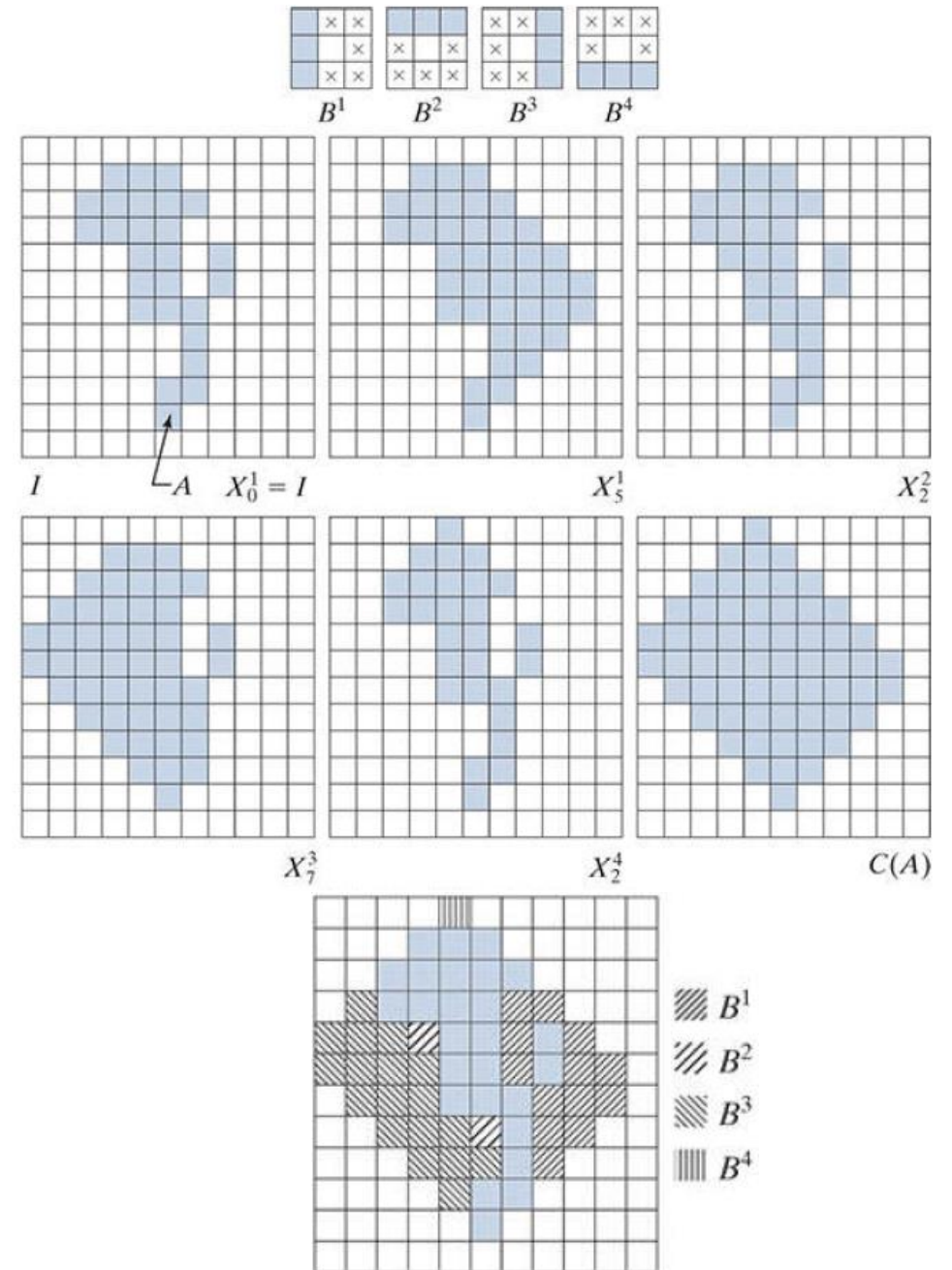
Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

Convex Hull

FIGURE 9.21

- (a) Structuring elements.
- (b) Set A.
- (c) – (f) Results of convergence with the structuring elements shown in (a).
- g) Convex hull.
- h) Convex hull showing the contribution of each structuring element.

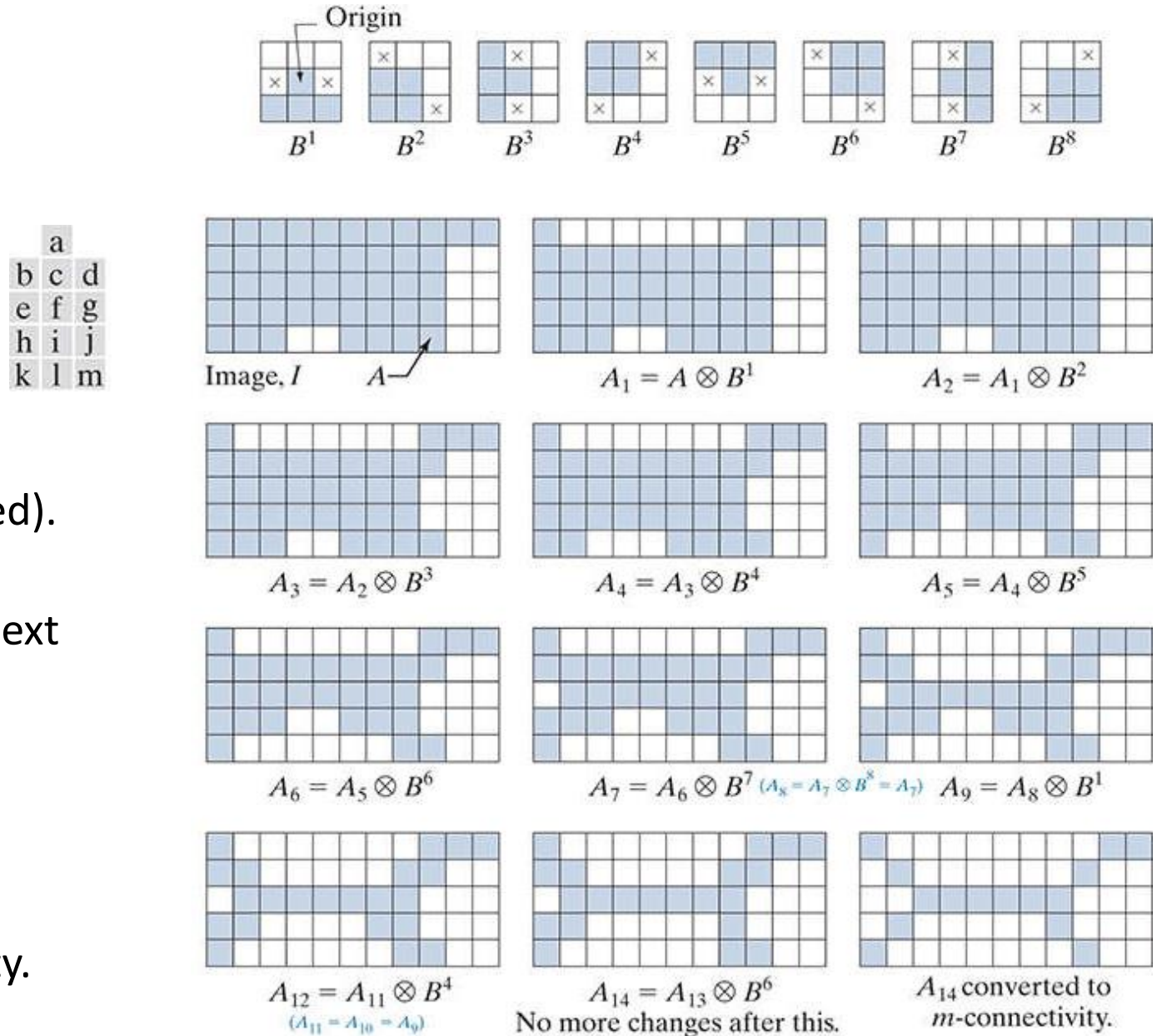
$$X_k^i = (X_{k-1}^i \circledast B^i) \cup X_{k-1}^i \quad i = 1, 2, 3, 4 \quad \text{and} \quad k = 1, 2, 3, \dots$$



Thinning

FIGURE 9.23

- Structuring elements.
- Set A.
- Result of thinning A with B^1 (shaded).
- Result of thinning A_1 with B^2
- – (i) Results of thinning with the next six SEs. (There was no change between A_7 and A_8)
- – (k) Result of using the first four elements again.
- Result after convergence.
- Result converted to m -connectivity.

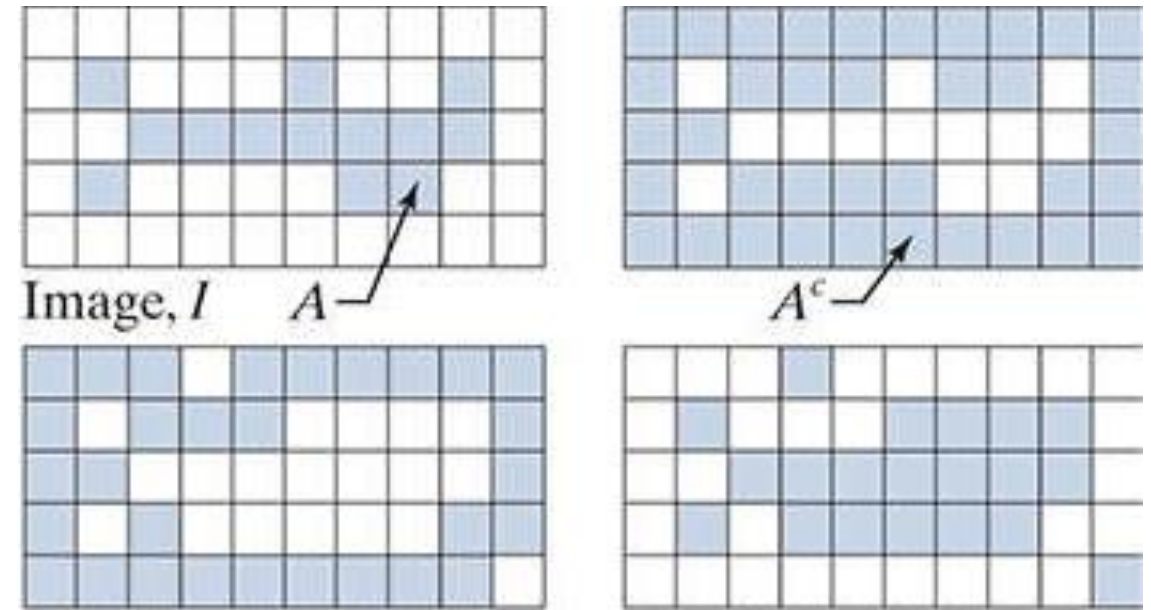


Thickening

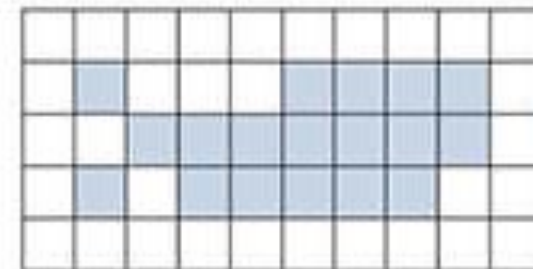
FIGURE 9.24

- (a) Set A.
- (b) Complement of A.
- (c) Result of thinning the complement.
- (d) Thickened set obtained by complementing (c).
- (e) Final result, with no disconnected points.

$$A \odot \{B\} = (((\dots((A \odot B^1) \odot B^2)\dots) \odot B^n)$$



a b
c d
e

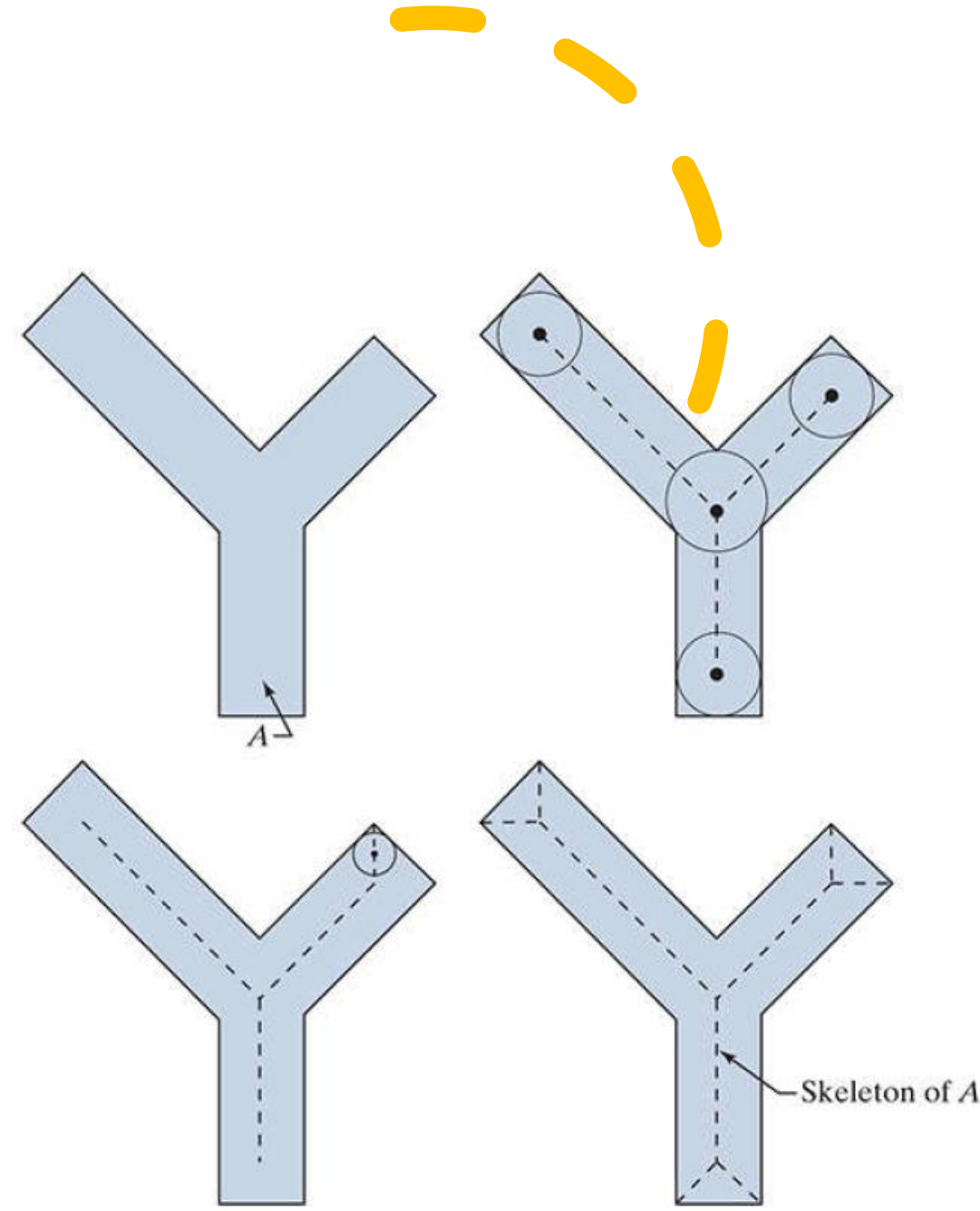


Skeleton

- A skeleton of a set A consists of points z that is the center of a maximum disk
- A maximum disk is a circle in A that can not be enclosed by another circle that is also in A .

FIGURE 9.25

- (a) Set A .
- (b) Various positions of maximum disks whose centers partially define the skeleton of A .
- (c) Another maximum disk, whose center defines a different segment of the skeleton of A .
- (d) Complete skeleton (dashed).



Skeleton equations

- Define k consecutive erosions of A as:

$$(A \ominus kB) = (((\dots((A \ominus B) \ominus B) \ominus \dots) \ominus B)$$

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B$$

- Let

$$K = \max\{k | (A \ominus kB) \neq \emptyset\}$$

- Then the skeleton can be found as:

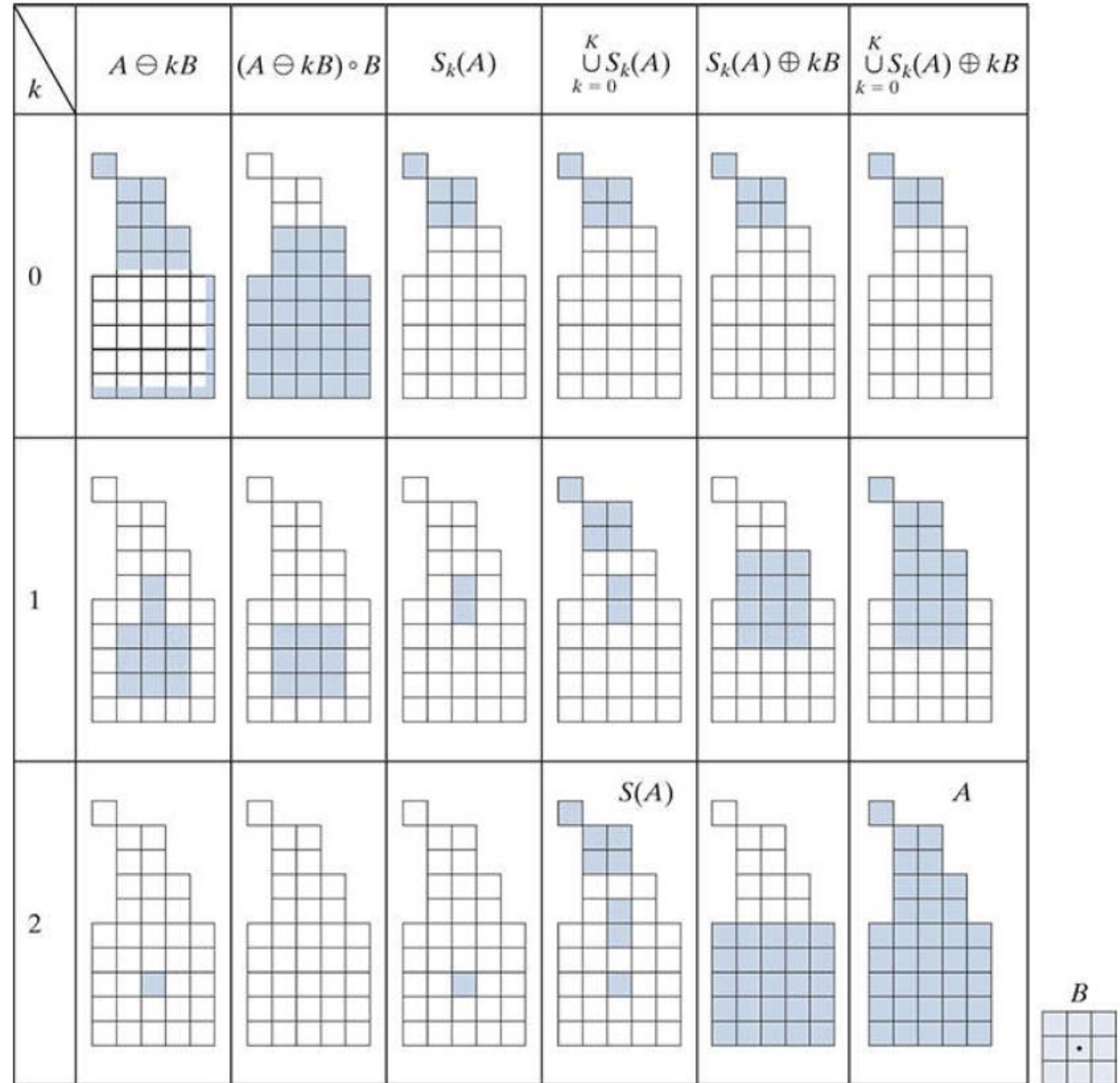
$$S(A) = \bigcup_{k=0}^K S_k(A)$$

$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

Skeleton computation

FIGURE 9.26

Implementation of **Eqs. (9-28)** through **(9-33)**. The original set is at the top left, and its morphological skeleton is at the bottom of the fourth column. The reconstructed set is at the bottom of the sixth column.

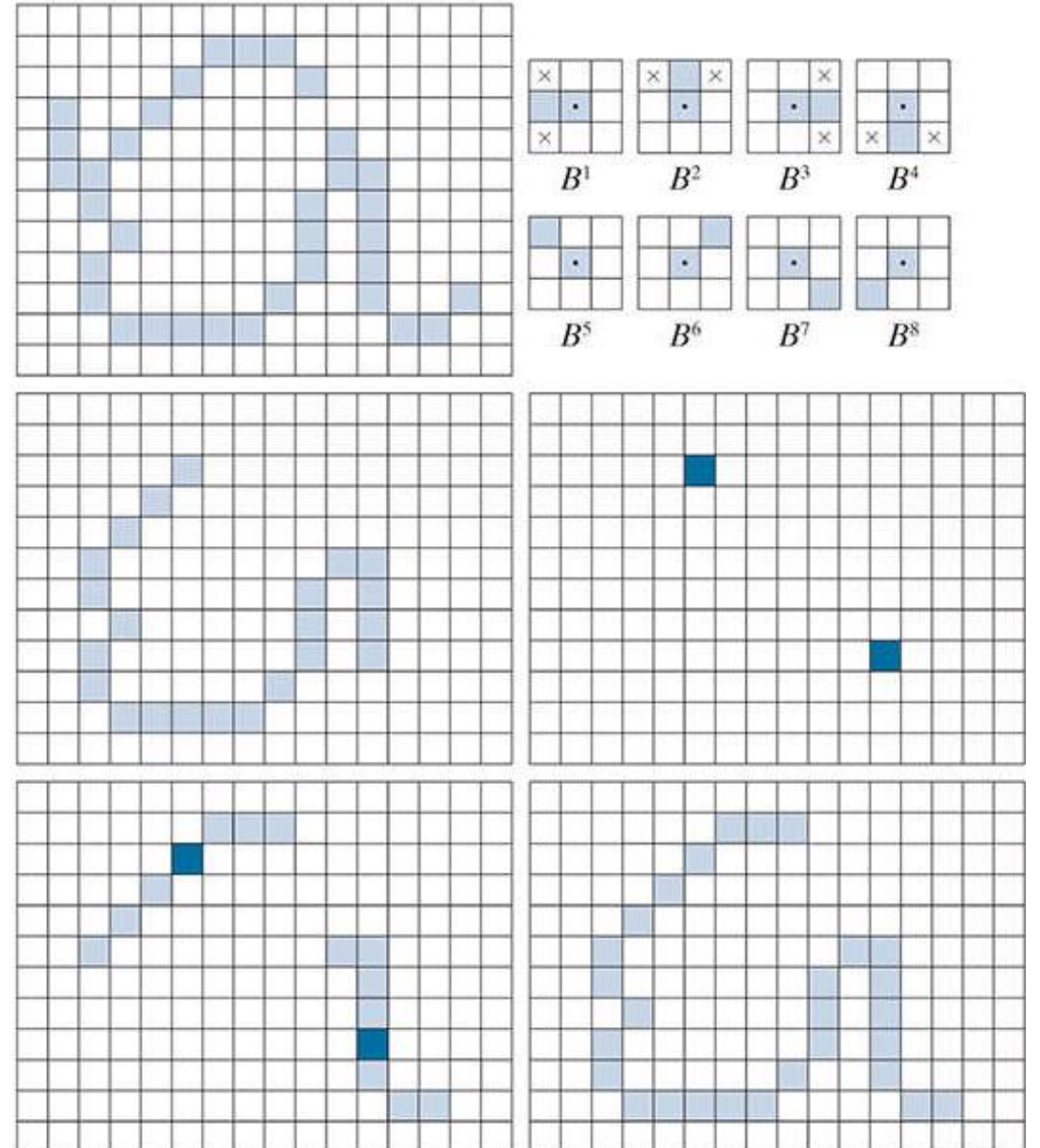


Pruning

FIGURE 9.27

- (a) Set A of foreground pixels (shaded).
- (b) SEs used for deleting end points.
- (c) Result of three cycles of thinning.
- (d) End points of (c).
- (e) Dilation of end points conditioned on (a).
- (f) Pruned image.

$$X_1 = A \otimes \{B\}$$



Terimakasih

